

 Marwadi University Marwadi Chandarana Group	Marwadi University Faculty of Engineering & Technology	
Subject: DSIP	Aim: Simulate cross correlation and autocorrelation on discrete time signals.	
Experiment:-3	Date:02-08-2026	Enrollment No:92400133074

Github <https://github.com/pateldhananjay39/DSIP.git>

➤ Objective:-

- To understand the mathematical definitions of autocorrelation and cross-correlation.
- To simulate these operations on deterministic and random discrete signals using Python.
- To interpret the results for signal similarity, periodicity, and time-delay estimation.

➤ Theory

- What is Correlation?

Correlation is a measure of similarity between two signals as a function of a time-lag applied to one of them. It is widely used in radar, sonar, communications (synchronization), and system identification.

- Autocorrelation

The autocorrelation function of a discrete-time signal $x[n]$ measures the similarity between the signal and a time-shifted version of itself. For a finite-length signal of length N , it is defined as:

$$r_{xx}[l] = \sum_{n=-\infty}^{\infty} x[n] \cdot x[n-l]$$

➤ where l is the lag (integer shift).

Interpretation:

- At $l = 0$, $r_{xx}[0] = \sum x[n]^2$, which is the total energy of the signal.
- Autocorrelation is even symmetric: $r_{xx}[l] = r_{xx}[-l]$.
- The maximum always occurs at zero lag: $|r_{xx}[l]| \leq r_{xx}[0]$.

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➤ Cross-Correlation:-

The cross-correlation between two different signals $x[n]$ and $y[n]$ measures how well $y[n]$ matches $x[n]$ when shifted:

$$r_{xy}[l] = \sum_{n=-\infty}^{\infty} x[n] \cdot y[n-l]$$

➤ Interpretation:

- If $y[n] = x[n-d]$ (i.e., a delayed version of x), then $r_{xy}[l]$ will have its peak at $l = d$.
- Cross-correlation is not generally symmetric.

➤ Normalized Correlation (Correlation Coefficient)

Often we normalize to get values between -1 and 1 for easier interpretation:

$$\rho_{xy}[l] = \frac{r_{xy}[l]}{\sqrt{r_{xx}[0] \cdot r_{yy}[0]}}$$

where $\rho = 1$ means identical signals (for that lag), $\rho = -1$ means opposite polarity, and $\rho = 0$ means completely unrelated.

➤ Applications in DSP

- Time-delay estimation: Finding the lag where cross-correlation peaks gives the time shift between two received signals (e.g., in target ranging).
- Periodicity detection: Autocorrelation of a periodic signal has peaks at multiples of the period.

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- Random signal analysis: White noise has an autocorrelation that is zero everywhere except at lag 0 (ideal for decorrelation).

1. Normalized cross-correlation:-

Code:-

```
import numpy as np
import matplotlib.pyplot as plt

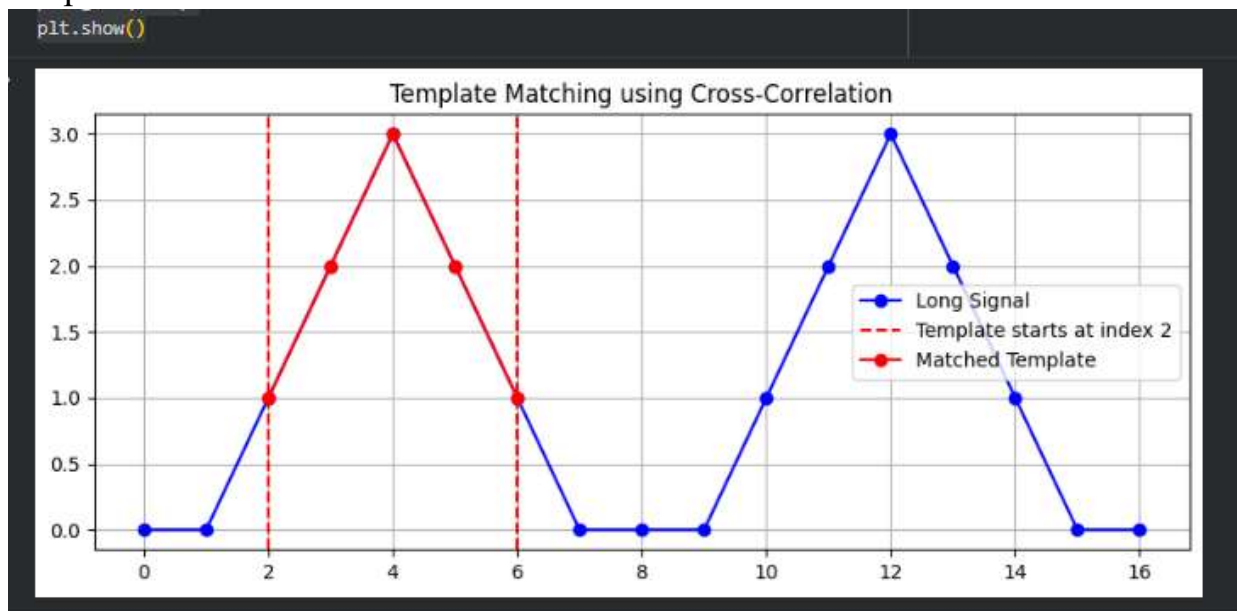
# Long signal
long_signal = np.array([0, 0, 1, 2, 3, 2, 1, 0, 0, 0, 1, 2, 3, 2, 1, 0, 0])
# Template to find
template = np.array([1, 2, 3, 2, 1])

# Compute cross-correlation
corr = np.correlate(long_signal, template, mode='valid')
peak_idx = np.argmax(corr)

plt.figure(figsize=(10, 4))
plt.plot(long_signal, 'bo-', label='Long Signal')
plt.axvline(peak_idx, color='red', linestyle='--', label=f'Template starts at index {peak_idx}')
plt.axvline(peak_idx + len(template) - 1, color='red', linestyle='--')
plt.plot(np.arange(peak_idx, peak_idx + len(template)), template, 'ro-',
label='Matched Template')
plt.legend()
plt.title('Template Matching using Cross-Correlation')
plt.grid(True)
plt.show()
```

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Output:-



2. Cross-Correlation to Detect Time Delay

Code:-

```
import numpy as np
import matplotlib.pyplot as plt

# Generate a random signal
np.random.seed(42)
x = np.random.randn(50)
# Create a delayed version (delay = 10 samples)
delay = 10
y = np.zeros_like(x)
y[delay:] = x[:-delay] # y[n] = x[n-10]
# Add a tiny bit of noise to make it realistic
y = y + 0.05 * np.random.randn(50)
```

 	Marwadi University Faculty of Engineering & Technology	
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Cross-correlation

```
cross_corr = np.correlate(x, y, mode='full')
```

```
# Lags for cross-correlation: length = len(x) + len(y) - 1 = 99
```

```
Nx = len(x)
```

```
Ny = len(y)
```

```
lags = np.arange(-Ny + 1, Nx)
```

```
# Find the peak lag
```

```
peak_idx = np.argmax(cross_corr)
```

```
estimated_delay = lags[peak_idx]
```

```
plt.figure(figsize=(12, 5))
```

```
plt.subplot(1, 2, 1)
```

```
plt.plot(x, 'b-', label='Original x[n]')
```

```
plt.plot(y, 'r--', label='Delayed y[n]')
```

```
plt.legend()
```

```
plt.title('Original and Delayed Signals')
```

```
plt.grid(True)
```

```
plt.subplot(1, 2, 2)
```

```
plt.stem(lags, cross_corr, basefmt="k-")
```

```
plt.axvline(estimated_delay, color='red', linestyle='--', label=f'Peak at lag = {estimated_delay}')
```

```
plt.title('Cross-Correlation rxy[l]')
```

```
plt.xlabel('Lag l')
```

```
plt.ylabel('Amplitude')
```

```
plt.legend()
```

```
plt.grid(True)
```

```
plt.tight_layout()
```

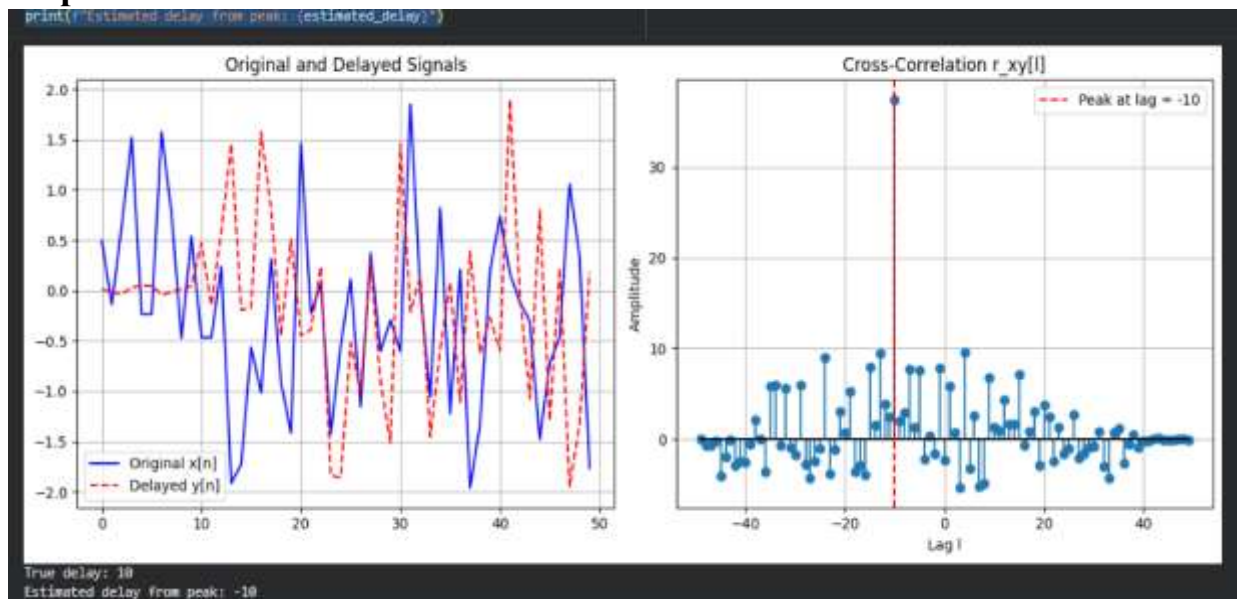
```
plt.show()
```

```
print(f"True delay: {delay}")
```

```
print(f"Estimated delay from peak: {estimated_delay}")
```

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output:-



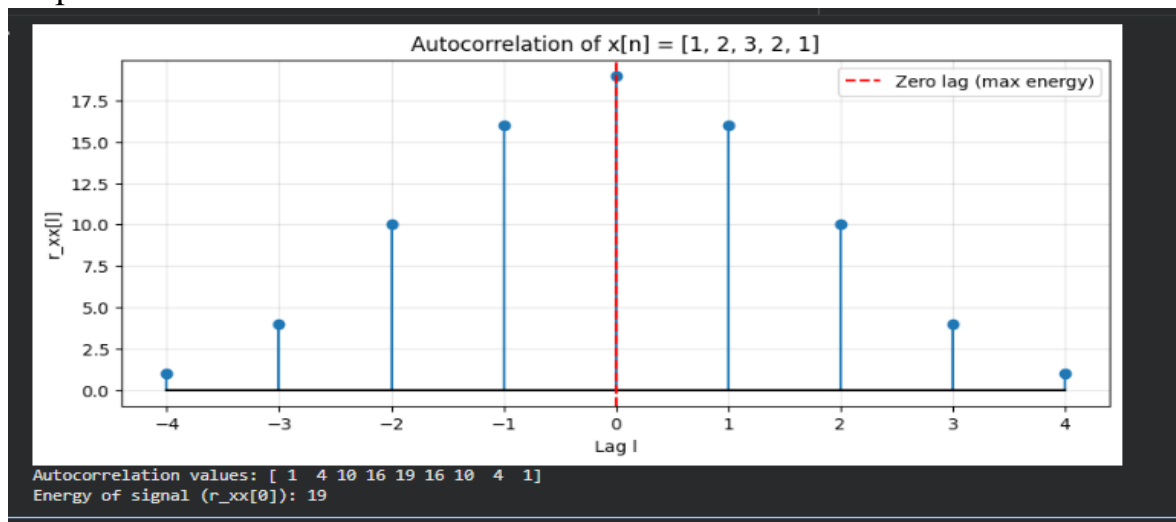
3. Autocorrelation of a Deterministic Signal

Code:-

```
import numpy as np
import matplotlib.pyplot as plt
# Define a simple deterministic signal
x = np.array([1, 2, 3, 2, 1])
# Compute autocorrelation using numpy.correlate (mode='full' preserves all lags)
auto_corr = np.correlate(x, x, mode='full')
# Create lag axis. For length N, lags go from -(N-1) to (N-1)
N = len(x)
lags = np.arange(-N + 1, N)
# Plot
plt.figure(figsize=(10, 4))
plt.stem(lags, auto_corr, basefmt="k-")
```

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```
plt.title('Autocorrelation of x[n] = [1, 2, 3, 2, 1]')
plt.xlabel('Lag l')
plt.ylabel('r_xx[l]')
plt.grid(True, alpha=0.3)
plt.axvline(0, color='red', linestyle='--', label='Zero lag (max energy)')
plt.legend()
plt.show()
print("Autocorrelation values:", auto_corr)
print("Energy of signal (r_xx[0]):", auto_corr[N-1])
output:-
```



4. Autocorrelation – Periodic vs. Random Signals

Code:-

```
import numpy as np
import matplotlib.pyplot as plt

n = np.arange(0, 200)
# Periodic signal: Sine wave
fs = 1
f = 0.05 # 5 cycles in 100 samples
```

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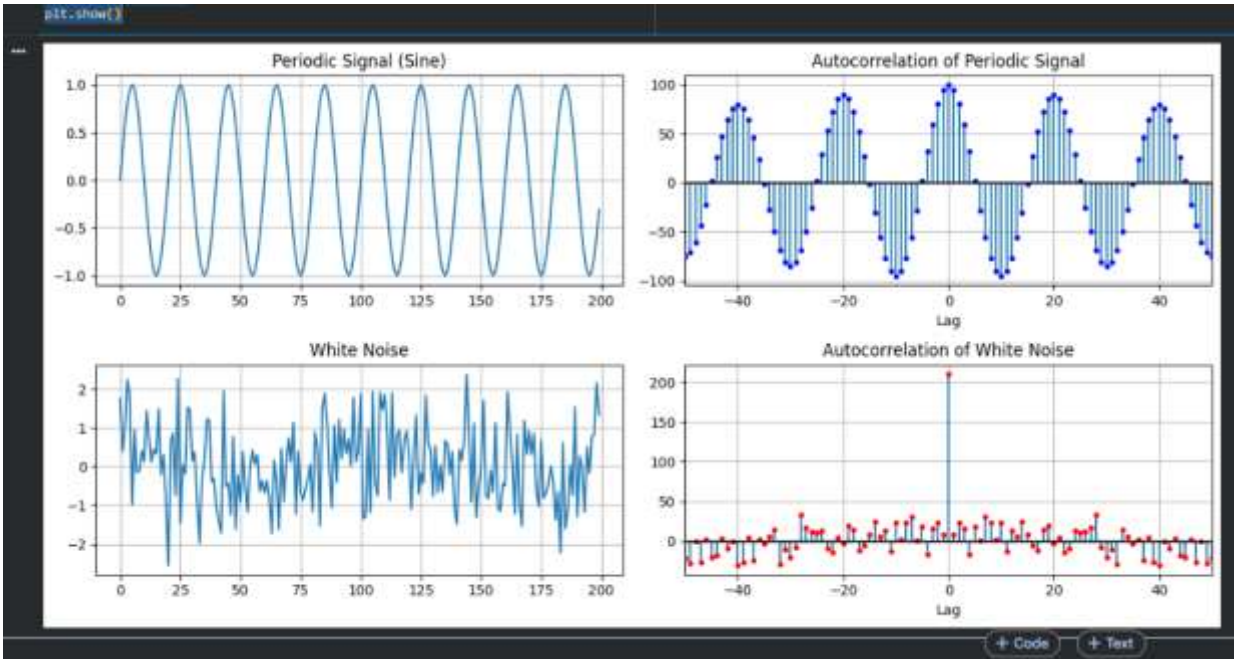
```
periodic_signal = np.sin(2 * np.pi * f * n)
```

```
# Random signal (White noise)
np.random.seed(0)
noise_signal = np.random.randn(200)
# Compute autocorrelations
auto_periodic = np.correlate(periodic_signal, periodic_signal, mode='full')
auto_noise = np.correlate(noise_signal, noise_signal, mode='full')
N = len(periodic_signal)
lags = np.arange(-N + 1, N)
plt.figure(figsize=(12, 6))
plt.subplot(2, 2, 1)
plt.plot(n, periodic_signal)
plt.title('Periodic Signal (Sine)')
plt.grid(True)
plt.subplot(2, 2, 2)
plt.stem(lags, auto_periodic, basefmt="k-", markerfmt='b.')
plt.title('Autocorrelation of Periodic Signal')
plt.xlabel('Lag')
plt.xlim(-50, 50)
plt.grid(True)
plt.subplot(2, 2, 3)
plt.plot(n, noise_signal)
plt.title('White Noise')
plt.grid(True)

plt.subplot(2, 2, 4)
plt.stem(lags, auto_noise, basefmt="k-", markerfmt='r.')
plt.title('Autocorrelation of White Noise')
plt.xlabel('Lag')
plt.xlim(-50, 50)
plt.grid(True)
```


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```
plt.tight_layout()
plt.show()
output:-
```



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Conclusion:-

- **Autocorrelation** confirmed symmetry and peaked at zero lag, accurately representing signal energy.
- **Cross-correlation** effectively estimated time delays, locating the peak at the correct shift even in the presence of noise.
- **Periodic and random signals** were clearly distinguishable—periodic signals showed sustained oscillatory correlations, while white noise produced an impulse-like response with near-zero values elsewhere.
- **Template matching** via cross-correlation successfully identified the position of a known pattern within a longer signal.